Volkan Çiçek

Senior AI Engineer & Data Scientist
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SUMMARY

Senior AI Engineer and Data Scientist with a Mathematics background and 5+ years at Logo Yazılım building production AI systems end to end. I work at both ends of the stack: designing and shipping LLM products (text-to-SQL, RAG, multi-agent systems) and owning the infrastructure they run on (Airflow/ETL, self-hosted GPU inference, MLOps). My text-to-SQL service runs in production across five ERP and HR products; the local LLM platform I built on NVIDIA DGX Spark serves 150+ users across the group; and through agent-driven engineering I ported 80–90% of a legacy UI migration in a few weeks, against two external bids of 5 and 12 months, before handing it to the owning team. MSc in Artificial Intelligence in progress.

EXPERIENCE

Senior Data Scientist
Sep 2025 – Present
Logo Yazılım
Gebze, Kocaeli (Hybrid)

- Own **PSA (Personalized SQL Agent)**, a production multi-tenant text-to-SQL service that turns natural language into SQL, NoSQL and function calls. Live across **five Logo ERP and HR products**: Tiger, Netsis, j-Platform, j-HR and Logo Lite. Built on FastAPI with async Logo PaaS integrations (the group's internal platform), automatic query correction, RAG retrieval over per-tenant schema and docs, prompt-token optimisation, per-tenant isolation, rate limiting and credit metering, and an **AST-level SQL/NoSQL validator** blocking write, DDL and injection patterns; hardened through load, concurrency and penetration testing plus production failure analysis.
- Designed, deployed and now operate the **group-wide local LLM inference platform** on an NVIDIA DGX Spark (GB10, 128 GB unified memory), serving **150+ users**: vLLM running an NVFP4-quantised coding model behind a **LiteLLM** gateway, plus Ollama and Open WebUI for team chat. Authenticated, per-user metered access for Claude Code, Continue and VS Code Copilot, with Prometheus/Grafana monitoring and automated alerting. Benchmarked at ~ **1,100 tok/s across 32 parallel users (+70% after tuning)**; offsets the company's paid Copilot licences and keeps source code inside the network.
- Proved out **agent-driven engineering** across the group with three deliveries:
  (1) migrated the UI of a UTM network-security appliance (Berqnet, Logo's cyber-security company) off a **331k-line, 2,500-file legacy Java/Vaadin** codebase onto a modern Nuxt/Vue frontend with a multi-agent LangGraph pipeline (orchestrator, explorer, coder), reaching **80–90% ported in a few weeks** against two external consultancy bids of 5 and 12 months, then handed to the owning team to finish;
  (2) built an **Agentic Development Platform PoC** to test whether the analyst team's agent-generated specs are code-ready: a multi-agent pipeline (plan / backend / frontend / integration / test / review) that produced a running Angular 17 + .NET 8 service against a live SOAP endpoint, with an auditable trace of every agent's input, output and open question as the actual deliverable;
  (3) built the calculation core of a deterministic payroll engine for Turkish social-security and tax authority (SGK/GİB) legislation (.NET/C#, 22-step flow, 13 payroll types, effective-dated legislation parameters, every calculation verified to the kuruş with an xUnit suite) **in one week**, scoped to the agreed rule set rather than a finished product, on a problem where three divergent in-house applications and the requirements analysis itself had failed to converge.
- Built the **evaluation harness** that gates changes to PSA: a hand-curated regression suite for each of the five products, an in-product **feedback loop** that grades answers against real user queries, and embedding A/B tests, so retrieval and prompt changes ship on measured accuracy, not intuition.
- Spearheaded a **centralised AI service architecture** for anomaly detection and forecasting, enabling cross-product integration and removing redundant per-product model deployments.
- **Internal reference point for AI across the company**: brief teams on where the field is moving, run knowledge transfer on agentic development tooling (Claude Code, Cursor, Copilot), and work with the AI Transformation Lead on adoption.
- **Mentor junior data scientists, interns and part-time staff (8 to date)**; set clean-code standards, own the team's AIOps and Generative AI roadmap, and turn business problems into deliverable projects with product and engineering counterparts.

Data Scientist
Jan 2022 – Sep 2025
Logo Yazılım
Kocaeli, Türkiye

- Core contributor to **Mariana**, a TÜBİTAK-funded (TEYDEB R&D grant) **AIOps** initiative: anomaly detection and forecasting over time-series telemetry from a **1,600-machine** estate, validated with hypothesis testing and RMSE/MAE monitoring to cut downtime.
- Engineered **ETL pipelines and Airflow orchestration** across two deployments for five cloud products (Logo Bulut ERP, Logo Bulut İK, Logo İşbaşı, Logo Lite, Logo Pera), powering multi-tenant licence, usage, sales and churn reporting sourced from REST APIs, PostgreSQL and web scrapers (Requests, BeautifulSoup, XPath).
- Led **Generative AI** delivery: intelligent chatbots, the first text-to-SQL engine, and Retrieval-Augmented Generation (RAG) systems for internal and customer-facing products.

- Built **AI code assistants and LLM-driven test-scenario generation** to shorten development cycles and raise software quality.
- Delivered **computer vision** in production: multi-model **OCR** pipelines for document digitisation integrated with Logo PaaS services (Logo Flow, Logo HR), and a deep-learning initiative analysing real-time video feeds to detect workplace-safety risks before incidents occur.
- Built a desktop **data-labelling application** (MongoDB + CSV pipelines) for non-technical teams, and drove **customer segmentation** (RFM) projects from ideation to production monitoring.

Junior Data Scientist — **Jul 2021 – Dec 2021**
Logo Yazılım — *Kocaeli, Türkiye*

- Optimised an **OCR** pipeline for Turkish character recognition (Python, OpenCV, Pillow), improving accuracy and throughput.
- Built a **FastAPI** service for the Polaris project, migrating Atlassian documentation into Logo DYS, the company's document management system, and using **BERT/T5** transformers to auto-generate a Turkish Q&A dataset that backed an Azure Knowledge Base chatbot.

Freelance Web Developer — **Jul 2015 – Jun 2021**
Self-employed — *Türkiye*

- Custom websites and e-commerce platforms (WordPress, OpenCart) for small businesses and student organisations, owned end to end from client requirements through to hosting.

EDUCATION

MSc, Artificial Intelligence (thesis, English) — **Sep 2025 – Jun 2027 (expected)**
Gebze Technical University — *Gebze, Türkiye*

BSc, Mathematics — **Sep 2018 – Jun 2022**
Gebze Technical University — *Gebze, Türkiye*
Thesis: *Principal Component Analysis* (advisor Prof. Dr. Nuri Çelik).

TECHNICAL SKILLS

Generative AI & Agents: LLM application design, multi-agent orchestration, RAG, text-to-SQL, function/tool calling, MCP servers, prompt engineering, evaluation and guardrails, embeddings, LangChain, LangGraph, LlamaIndex, HuggingFace Transformers
Models & Platforms: Anthropic Claude, OpenAI, Azure OpenAI, Google Gemini, open-weight LLMs and vision-language models served locally
Machine Learning: Deep learning, time-series forecasting, anomaly detection, NLP, computer vision (OCR, VLMs), exploratory data analysis, customer segmentation (RFM), A/B testing, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, OpenCV
Languages & Frameworks: Python, SQL, C#/.NET, FastAPI, pytest, JavaScript/Vue/Nuxt, Bash, Git
Data & Infrastructure: Apache Airflow, ETL/ELT, PostgreSQL, MongoDB, DuckDB, vector databases (pgvector, Milvus), REST API design, microservices, multi-tenant architecture
MLOps & Platform: Docker, Kubernetes, CI/CD (Jenkins), MLflow, Prometheus + Grafana, vLLM, LiteLLM, Ollama, inference cost and token optimisation, Azure ML Studio, Proxmox virtualisation and self-hosted Linux infrastructure
Ways of working: Technical leadership, mentoring, Agile/Scrum, clean code and code review, technical writing (EN/TR)
Languages: Turkish (native), English (full professional proficiency)

WRITING & OPEN SOURCE

- Technical blog at volkancicek.com, where I publish long-form analysis of agentic engineering, LLM inference infrastructure and legacy-code modernisation (e.g. *"Where Does an Agent Live?"*, *"What Can We Do With an NVIDIA DGX Spark?"*).
- `litellm-supply-chain-audit`: an open-source CLI that statically detects the compromised LiteLLM PyPI releases from the March 2026 supply-chain attack, without importing the package.
- VoiceDub Studio, a self-built AI dubbing and voice-cloning product (TypeScript): designed, built and deployed solo, from product concept through to a live site.
- Upstream contributions to the tools I run in production. Diagnosed and reported a memory leak in `doctr` under FastAPI, an arm64 image shipping an amd64 binary in `nvidia_gpu_exporter`, and a version-bump gap in the GB10 vLLM build; merged pull requests to `daikin_onecta` and `HostsFileEditor`.

CERTIFICATIONS & ADDITIONAL

- **Machine Learning Winter School**, Koç University & İş Bankası · **Mathematics for Machine Learning**, Nesin Mathematics Village · **NVIDIA** Fundamentals of Deep Learning · **BT Akademi** DevOps, Model Deployment, ML Model Validation, Container Architecture & Kubernetes
- **Chairman of the Board**, GTU Mathematics Club · **Financial Director**, IEEE GTU Student Branch · **Organisation Chief**, 9th Science & Technology Days (100-person team, 30 sponsors, 21 speakers)